MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular Examination October 2025
Third Semester
Discipline Specific Course (DSC) -MG3DSCECO201 – QUANTITATIVE ECONOMICS -I
(2024 ADMISSION ONWARDS)

Duration : 2 hours **Maximum Marks: 70**

Scheme for valuation

Part A

Multiple Choice Questions

Answer all questions.

Each question carries 1 mark.

Give 1 mark each to the correct answers of Qn 1-10.

1. a) r x s elements
2. c) 8 outcomes
3. d) C = a+bY
4. b) $12x^2$
5. a) Qs=f(p)
6. b) 5
7. a) Every element of the domain has exactly one image in the range
8. d) 3
9. a) $\begin{bmatrix} 6 & -6 \\ -2 & 4 \\ -4 & 8 \end{bmatrix}$
10. d) f'(x) + g'(x)

Part B

Short Answer

Answer any 10 questions.

Each question carries 3 marks.

11. a) BUC= {1,3,5,6,7,8} – ½ **mark**
AU(BUC)={1,2,3,4,5,6,7,8} – **1 mark**

b) AUB={1,2,3,4,57} – ½ **mark**

(AUB)∩C= {1,5} – **1 mark**

12. ln (6) = ln (3x2) – **1 mark**
=ln 3+ ln 2 – **1 mark**
=x+y **- 1 mark**

13. Definition of square matrix : A **square matrix** is a matrix that has the **same number of rows and columns**. – 1.5 marks
Definition of rectangular matrix :: A **rectangular matrix** is a matrix that has a **different number of rows and columns** – 1.5 marks
Or
One example each for both – 1 mark each

Maximum 3 marks

14. f'(x)= 6x-12 – **1 mark**
f'(x)=0 implies 6x – 12 = 0, solving gives x=2 – **1 mark**
f"(x) = 6 > 0 implies function is minimum at x=2 – **1 mark**
Or
Only the two conditions of minimum according to differentiation – **1 mark**
Maximum 3 marks

15. Definition A **simple event** is an event that **contains only one sample point as its elements** – **2 marks. Example:** When tossing a coin: getting head is a simple event, getting tail is another simple event. When rolling a die: getting **3** is a simple event, getting **6** is also a simple event. **Symbolically:** If the sample space S={1,2,3,4,5,6} then E={3} is a simple event. – **1 mark**

Or
Example alone – **2 marks**
Maximum 3 marks

16. $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ **- 1.5 marks**
$(A^T)^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ = A – **1.5 marks**

17. MR = $\frac{dR}{dQ}$ **- 1 mark**
= 100-4Q – **2 marks**

18. Statement P(AUB) = P(A)+P(B), if A and B are mutually exclusive events – **2 marks**
Illustration – One example – **1 mark**

19. Ax = b, $\begin{bmatrix} 2 & -1 \\ 7 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -7 \end{bmatrix}$ **- 1 mark**
|A| = 2x-2 - -1x7 = -4 + 7 =3 **– 1 mark**
x=$\frac{|A1|}{|A|}$= -9/3 = - 3 – ½ **mark**
y= $\frac{|A2|}{|A|}$=-21/3 = - 7 – ½ **mark**

20. Give **3 marks to correct answer of either of the two.**

(i) Minor = $\begin{bmatrix} 4 & 3 \\ 2 & 2 \end{bmatrix}$ **– 1 mark**
Cofactor = $\begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ Adj (A) = $\begin{bmatrix} 4 & -2 \\ -3 & 2 \end{bmatrix}$ **- 1 mark**
|A| = $\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = 8 - 3 = 5$
$A^{-1} = \frac{Adj\,(A)}{|A|} = \begin{bmatrix} 4/5 & -2/5 \\ -3/5 & 2/5 \end{bmatrix}$ **- 1 mark**
Or

(ii) Minor = $\begin{bmatrix} 3 & -2 \\ -2 & 5 \end{bmatrix}$ **- 1 mark**
Cofactor = $\begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix}$ Adj (B) = $\begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix}$ **- 1 mark**
|B| = $\begin{vmatrix} 5 & 7 \\ -2 & 3 \end{vmatrix} = 15 - -14 = 29$
$B^{-1} = \frac{Adj\,(B)}{|B|} = \begin{bmatrix} 3/29 & 2/29 \\ 2/29 & 5/29 \end{bmatrix}$ **- 1 mark**

Maximum 3 marks

21. $f(x) = y^4$ where $y=(3x^2+2)$ **– 1 mark**

$f'(x) = 4y^3.(6x+0) = 24x\ y^3 = 24x(3x^2+2)^3$ – **2 marks**

Or

Chain rule of differentiation only – **1 mark**

Maximum 3 marks

22. A – B = $\begin{bmatrix} 4 & -4 \\ -1 & 5 \end{bmatrix}$ **- 1 ½ marks**

B – A = $\begin{bmatrix} -4 & 4 \\ 1 & -5 \end{bmatrix}$ **- 1 ½ marks**

23. Definition of revenue function in terms of quantity sold and price at which it sold – **3 marks**

24. Statement P(AUB) = P(A)+P(B), if A and B are mutually exclusive events – **1 ½ marks**

P(AUB) = P(A)+P(B) – P(A ∩ B), for any two events – **1 ½ marks**

25. Definition : The **degree of an equation** is the **highest power of the variable** in the equation - – **2 marks**

Examples: Linear equation: $2x+3=0$ → Highest power of x is **1**, so the **degree = 1**. **Quadratic equation:** $x^2+5x+6=0$→ Highest power of x is **2**, so the **degree = 2**. **Cubic equation:** $x^3-4x+2=0$ → Highest power of x is **3**, so the **degree = 3**. – **1 mark**

Part C

Short Essay

Answer any 5 questions.

Each question carries 6 marks.

26. Ans: No, it is not continuous at x=1 – **2 Marks**

Reason – At x=1, the function is not defined as division by zero happens. So it is not continuous – **4 Marks**

Or

For finding the limits (which took the form 0/0) only or any reasonable attempt– **2 marks**

Maximum 6 marks

27. Error occurs in the given consumption function of the question C= $\bar{C}$+ cY, . So give **full marks (6) for any reasonable attempt.**

ANSWER; The equilibrium level of income (Y) is determined where consumption equals investment.

Given: - Consumption function: C = $\bar{C}$+ cY, where $\bar{C}$ = 40 and c = 0.7

- Investment function: I = I* = 80

Y = C + I = ($\bar{C}$ + cY) + I* = 40 + 0.7Y + 80 = 120 + 0.7Y

At equilibrium, Y = 120 + 0.7Y

To solve for Y, subtract 0.7Y from both sides:

0.3Y = 120, Y = 120 / 0.3, Y = 400

The equilibrium level of income is 400. Value Liberally

28. A×B= $\begin{bmatrix} 52 & 54 & 38 \\ -11 & -30 & -100 \\ 46 & 4 & 14 \end{bmatrix}$ – **3 marks**

$B \times A = \begin{bmatrix} 71 & 23 & 69 \\ -10 & -80 & -78 \\ -23 & 29 & 45 \end{bmatrix}$ – **3 marks**

29. a. Ans: $6x^5+12x^2-7$ – **2 marks**

b. Ans: $5x^4-12x^3+4x-1$ – **2 marks**

c. Ans: $9x^8-28x^6+15x^2-1$ – **2 marks**

Or Statement of power rule of differentiation **only – 3 marks**

Maximum 6 marks

30. Answer **any 3 completely – 2 marks each** – bound to 6 Marks (**Maximum 6 marks**)

1. Apply Product rule. Ans: $(x^2+3x)(3x^2)+(x^3+2)(2x+3)$

2. Apply Quotient rule. Ans: $\frac{x.2x-(x^2+1)}{x^2} = \frac{2x^2-x^2-1}{x^2} = \frac{x^2-1}{x^2}$

3. Apply Quotient rule. Ans: $\frac{(x+2).3-(3x-5).1}{(x+2)^2} = \frac{11}{(x+2)^2}$

4. Apply Chain rule. $y=z^{-4}$; $z=(2x^2+1)$

Ans: y'= $-4y^{-5}.(4x+0)= -4(2x^2+1)^{-5}.4x= -16x(2x^2+1)^{-5}$

5. Apply Quotient rule. Ans: $y=\log z$; $z=(x^2+4)$

Ans: y'= $\frac{1}{z}.(2x+0) = \frac{2x}{(x^2+4)}$

6. $y'=7.(3.e^{3x})=21e^{3x}$

31. Random experiment – definition – **1 mark**

Example – **1 mark**

Trial – definition **– 1 mark**

Example – **1 mark**

Outcome – definition – **1 mark**

Example – **1 mark**

[A random experiment is a process or action that can be repeated under the same conditions, but the outcome cannot be predicted with certainty in advance.Example:

Tossing a coin — you cannot be sure whether you'll get a Head or a Tail.A trial is a single performance or repetition of a random experiment.Example:If you toss a coin once, that single toss is one trial. Tossing it three times means three trials.An outcomeis the possible result of a random experiment. Example**:** When you toss a coin, the possible outcomes are Head (H) or Tail (T)**.]**

32. $|A|=1\begin{vmatrix} 5 & -1 \\ 1 & 2 \end{vmatrix} - 0.\begin{vmatrix} 3 & -1 \\ 0 & 2 \end{vmatrix} + 4\begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix}$ **- 3 Marks**

=1.(10+1) – 0.(6-0)+4.(3-0)

=11-0+12

=23 **- 3 Marks**

33. Step 1: Meaning **(2 marks)**

- *Total concepts* measure the overall amount (e.g., Total Cost, Revenue, Utility).
- *Marginal concepts* measure the additional change in total when one more unit is produced or consumed.

Step 2: Examples and Derivation **(2 marks)**

- Cost: MC = ΔTC/ΔQ → if TC rises from 500 to 520, MC = 20
- Revenue: MR = ΔTR/ΔQ → if TR rises from 1000 to 1070, MR = 70
- Utility: MU = ΔTU/ΔQ → if TU rises from 30 to 36, MU = 6
- In calculus: M=d(T)/dQ (e.g., if $TC=100+5Q+Q^2$, then MC=5+2Q)

Step 3: Importance in Decision-Making **(2 marks)**

- Firms maximize profit where MC = MR.

- Consumers reach equilibrium where MU = Price.
- Marginal analysis guides optimal production, pricing, and resource use.

Total = **6 Marks (2 + 2 + 2)**

34. When A and B are dependent events, the probability of A given B is given by the formula: $P(A|B)=P(A\cap B)/P(B)$ Alternatively, $P(B|A)=P(A\cap B)/P(A)$

Since A and B are dependent, we also know that:$P(A\cap B)=P(A|B)\cdot P(B)=P(B|A)\cdot P(A)$

Hence, dependence means that $P(A|B)\neq P(A)$ and $P(B|A)\neq P(B)$.

Such **proper explanation - 6 Marks**

35. $B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & -1 \\ 5 & 2 & 2 \end{bmatrix}$

Cofactor $C_{ij} = (-1)^{(i+j)} * M_{ij,}$

where M_{ij} is the minor – determinant obtained by deleting the i-th row and j-th column.

$M_{11} = \begin{vmatrix} 4 & -1 \\ 2 & 2 \end{vmatrix}$ = (4)(2) - (-1)(2) = 10, C_{11} = +10

$M_{12} = \begin{vmatrix} 3 & -1 \\ 5 & 2 \end{vmatrix}$ = (3)(2) - (-1)(5) = 11, C_{12} = -11

$M_{13} = \begin{vmatrix} 3 & 4 \\ 5 & 2 \end{vmatrix}$ = (3)(2) - (4)(5) = -14, C_{13} = -14

$M_{21} = \begin{vmatrix} 0 & 1 \\ 2 & 2 \end{vmatrix}$ = (0)(2) - (1)(2) = -2, C_{21} = 2

$M_{22} = \begin{vmatrix} 2 & 1 \\ 5 & 2 \end{vmatrix}$ = (2)(2) - (1)(5) = -1, C_{22} = -1

$M_{23} = \begin{vmatrix} 2 & 0 \\ 5 & 2 \end{vmatrix}$ = (2)(2) - (0)(5) = 4, C_{23} = 4

$M_{31} = \begin{vmatrix} 0 & 1 \\ 4 & -1 \end{vmatrix}$ = (0)(-1) - (1)(4) = -4, C_{31} = -4

$M_{32} = \begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix}$ = (2)(-1) - (1)(3) = -5, C_{32} = +5

$M_{33} = \begin{vmatrix} 2 & 0 \\ 3 & 4 \end{vmatrix}$ = (2)(4) - (0)(3) = 8, C_{33} = +8

Minors or Matrix of minors – **4 Marks**

$C = \begin{bmatrix} 10 & -11 & -14 \\ 2 & -1 & -4 \\ -4 & 5 & 8 \end{bmatrix}$ – **1 Mark**

The adjoint of a matrix is the transpose of its cofactor matrix.

$Adj(B) = C^T = \begin{bmatrix} 10 & 2 & -4 \\ -11 & -1 & 5 \\ -14 & -4 & 8 \end{bmatrix}$ **- 1 Mark**